

Question 1:

What is CSS Flexbox, and how is it useful for layout design? Explain the terms flex-container and flex-item.

➤ **CSS Flexbox**

- **Flexbox (Flexible Box Layout)** is a CSS layout model used to arrange elements easily and efficiently.
- It helps create responsive and flexible layouts without using float or positioning.

➤ **Usefulness:**

- Automatically adjusts items to fit different screen sizes.
- Makes alignment, spacing, and distribution of items simple.
- Useful for horizontal and vertical centering of elements.

➤ **Key Terms:**

→ **Flex Container:**

- The parent element with display: flex: .
- Controls the layout and alignment of its child items.

→ **Flex Items:**

- The child elements inside a flex container.
- These items are arranged and controlled by Flexbox properties like justify-content, align-items, and flex-wrap.

Question 2:

Describe the properties justify-content, align-items, and flex-direction used in Flexbox.

➤ **Flexbox Properties**

➤ **justify-content:**

- Aligns flex items horizontally (along the main axis).
- Common values:
 - flex-start → items start from left
 - center → items centered horizontally
 - flex-end → items move to right

- space-between → equal space between items
- space-around → equal space around items

➤ **align-items:**

→ Aligns flex items vertically (along the cross axis).

→ Common values:

- flex-start → items at top
- center → items centered vertically
- flex-end → items at bottom
- stretch → items fill container height

➤ **flex-direction:**

→ Defines the direction of the main axis (how items are placed).

→ Common values:

- row → items in a horizontal row (default)
- row-reverse → items in reverse row order
- column → items in a vertical column
- column-reverse → items in reverse column order.